

Sri Lanka Institute of Advanced Technological Education



Higher National Diploma in Information Technology

Advanced Technological Institute, Dehiwala

Individual Project Proposal

"Student Mental Health Support System"

Student Name: Dishmi Dahara Gajanayaka

Registration Number: DEH/IT/2324/F/0044

Statement of Originality

I hereby declare that this project titled "Student Mental Health Support System" is my own work and has been carried out under the supervision of the project supervisor.

This project proposal and the proposed system are the result of my original ideas, research, and efforts. All information, diagrams, and content obtained from other sources have been properly acknowledged and referenced where necessary.

I confirm that this work has not been submitted in whole or in part for any other academic qualification or degree at this or any other institution.

I understand that any form of plagiarism will result in appropriate disciplinary action.

Acknowledgement

I would like to express my sincere gratitude to my project supervisor for their continuous guidance, support, and valuable feedback throughout the development of this project. Their insights and encouragement have greatly contributed to the successful completion of this proposal.

I also wish to thank the institution for providing the necessary resources and environment to carry out this project.

Furthermore, I would like to extend my appreciation to my friends and family for their support, motivation, and encouragement during this process.

Contents

Statement of Originality.....	2
Acknowledgement	3
1.Project Title.....	5
2.Introduction.....	5
3. Problem Statement	6
Who faces the problem?.....	6
What is the problem?	6
Why does it need to be solved?.....	6
4. Objectives.....	7
Main Objective.....	7
Specific Objectives	7
5. Scope of the System.....	8
In Scope.....	8
Out of Scope.....	8
6. Proposed Solution	9
7. System Features	10
7.1 User Dashboard Features	10
7.2 Admin Dashboard Features.....	11
8. Methodology	12
9. Technologies to be Used	13
10. Expected Outcomes.....	14
11. Target Users	15
12. Timeline	16
13. Conclusion	17

1.Project Title

"Student Mental Health Support System"

2.Introduction

The Student Mental Health Support System is a web/mobile-based platform designed to support the mental well-being of students. In today's fast-paced academic environment, students often face stress, anxiety, and depression due to academic pressure, personal issues, and social challenges. However, access to proper mental health support is limited, and many students hesitate to seek help due to stigma or lack of awareness.

This system aims to provide a safe, private, and accessible platform where students can track their mood, book counselling sessions, communicate anonymously, and access mental health resources. The integration of AI-based features further enhances user support by providing instant guidance and assistance.

The updated system introduces a two-tier dashboard architecture, a User Dashboard and an Admin Dashboard enabling personalized, real-time support through two-way communication, mood-based activity recommendations, and comprehensive administrative tools. These enhancements are designed to improve student engagement and deliver more effective mental health interventions.

3. Problem Statement

Who faces the problem?

- School and university students
- Teenagers facing academic and personal stress

What is the problem?

- Increasing mental health issues such as stress, anxiety, and depression
- Limited access to counselling services
- Hesitation to seek help due to stigma
- Lack of tools for self-monitoring mental health
- No personalized activity or coping strategy recommendations
- Absence of real-time two-way communication between students and support staff

Why does it need to be solved?

- Mental health directly impacts academic performance and daily life
- Early intervention can prevent serious mental health conditions
- A secure digital platform can encourage students to seek help easily and privately
- Personalized recommendations improve the effectiveness of mental health interventions

4. Objectives

Main Objective

- To design and develop a secure, user-friendly, and feature-rich system to support student mental health through personalized support, two-way communication, and AI-powered recommendations.

Specific Objectives

1. To develop a comprehensive User Dashboard with mood tracking, resources, and activity features
2. To implement a full-featured Admin Dashboard for user management, analytics, and content control
3. To implement a counselling session booking system
4. To create a two-way communication system between users and administrators
5. To create an anonymous chat feature for peer and support communication
6. To enhance the Mood Tracker with mood-based activity and game recommendations
7. To provide curated mental health resources such as articles and tips
8. To develop an AI chatbot for basic mental health assistance
9. To implement a notification and emergency alert system
10. To enable administrators to analyze mood records and recommend personalized activities

5. Scope of the System

In Scope

1. User registration
2. User Dashboard (mood tracking, resources, games, communication)
3. Admin Dashboard (user management, analytics, content, reports)
4. Mood tracking dashboard with daily mood logging
5. Mood-based game and activity recommendation engine
6. Two-way messaging between users and administrators
7. Counsellor booking system
8. Anonymous peer chat system
9. AI chatbot support
10. Notifications, reminders, and emergency alerts
11. Mental health resources section
12. System usage statistics and report generation

Out of Scope

1. Live video/audio counselling sessions
2. Clinical diagnosis or prescription of medication
3. Integration with external hospital or healthcare systems

6. Proposed Solution

The proposed Student Mental Health Support System is a web-based application designed to support students in managing their mental well-being through digital tools and resources. The system provides the following functionalities:

1. User Registration and Login

Students can create accounts and securely access the system using authentication features.

2. Mood Tracking System

Users can record their daily moods and view their mood history to better understand their emotional well-being over time.

3. Mental Health Resources

The system provides articles, tips, and educational content related to mental health awareness and stress management.

4. Counselling Session Booking

Students can schedule appointments with counsellors through the system.

5. Anonymous Chat Support

Users can communicate anonymously with peers or support staff, encouraging open discussions without revealing their identity.

6. AI Chatbot Assistance

An AI-powered chatbot provides instant responses, guidance, and basic mental health support at any time.

7. Activity and Game Recommendations

The system recommends suitable relaxation activities and stress-relief games based on the user's current mood.

8. Communication System

Students can send messages, feedback, and support requests directly to administrators or counsellors.

9. Notification and Reminder System

Users receive reminders for counselling sessions, mood tracking activities, and important announcements.

10. Admin Management and Reporting

Administrators can manage users, monitor mood records, generate reports, and provide personalized support to students.

7. System Features

7.1 User Dashboard Features

Feature	Description
1. User Registration & Login	Secure sign-up and authentication using JWT
2. Mood Tracker	Users can record their daily mood, view mood history and trends, maintain journal entries, and receive personalized activities, relaxation games recommendations based on their mood.
3. Mental Health Resources	Access to curated articles, tips, guided meditations, and exercises.
4. Counselling Booking	Book sessions with available counsellors
5. Anonymous Chat	Communicate with peers or support staff without revealing identity
6. AI Chatbot	24/7 AI-powered chatbot for immediate mental health guidance
7. Messaging to Admin	Send messages, feedback, or support requests directly to administrators
8. Notification Center	Receive reminders, session alerts, and admin responses

7.2 Admin Dashboard Features

Feature	Description
1. User Management	View, edit, activate, or deactivate student and counsellor accounts
2. Mood Analytics & Reporting	Analyze individual and aggregate mood records; generate trend reports
3. Content Management	Add, edit, and remove mental health resources, articles, and tips
4. Recommendation Management	Curate and assign games/activities to individual users based on mood data
5. Communication Management	Respond to user messages, support requests, and provide personalized guidance.
6. Feedback & Complaint Handling	Review and resolve user-submitted feedback and complaints
7. Notification & Announcement Management	Send system-wide or targeted notifications and announcements
8. User Activity Monitoring	Track login frequency, mood entries, and resource usage patterns
9. Chat & Support Management	Oversee anonymous chat sessions and moderate content
10. Emergency Support Alerts	Receive and respond to high-risk user alerts escalate when necessary
11. System Usage Statistics	Dashboard overview of active users, session counts, and engagement metrics
12. Report Generation	Export mood trends, session statistics, and system activity reports

8. Methodology

The system will be developed using the Agile development methodology to ensure flexibility, continuous improvement, and iterative feedback integration. The development process includes:

- Requirement gathering
- System design (UI/UX and database architecture)
- Iterative implementation in sprints
- Continuous testing and debugging
- Deployment and finalization

Each phase will be carried out iteratively, allowing feedback and improvements throughout the project lifecycle. Regular sprint reviews will ensure that new features, especially the admin dashboard enhancements and recommendation engine meet stakeholder expectations.

9. Technologies to be Used

Category	Technology / Tool
Frontend	React.js / HTML, CSS, JavaScript
Backend	Node.js with Express.js
Database	MongoDB
AI Integration	API-based or rule-based chatbot; recommendation logic
Authentication	JSON Web Tokens (JWT)
Real-Time Communication	Socket.IO (for two-way messaging and live notifications)
Charts & Analytics	Chart.js or Recharts (for mood trend visualizations)
Version Control	GitHub
Development Tools	Visual Studio Code

10. Expected Outcomes

- A comprehensive digital platform for student mental health support
- Improved stress management and emotional awareness through daily mood tracking
- Easy and private access to counselling services
- An anonymous peer support system for students
- AI-based quick mental health assistance available 24/7
- Personalized activity and game recommendations based on individual mood states
- Efficient admin tools for monitoring, analyzing, and responding to student mental health data
- Enhanced overall student well-being through timely and personalized interventions
- Streamlined emergency support for high-risk students

11. Target Users

Primary users	Specialist users
School Students	Counsellors
University Students	Educational Institutions
Teenagers	Psychologists
Parents	Academic Advisors / Mentors

12. Timeline

Sprint	Phase	Duration	Key Activities
Sprint 0	Preparation Phase	Week 1-4	Proposal preparation, requirement gathering, system design (UI/UX, database, dashboard architecture)
Sprint 1	Core Development	Week 5-10	User & Admin Dashboard development, Mood Tracker, Two-Way Communication, AI Chatbot, Booking System
Sprint 2	Testing	Week 11-13	Testing all features, debugging, UAT, recommendation engine validation, performance improvements
Sprint 3	Finalization	Week 14-15	Final refinements, report generation features, project submission and documentation

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Preparation Phase															
2. Core Development															
3. Testing															
4. Finalization															

13. Conclusion

The Student Mental Health Support System provides a modern, comprehensive, and effective solution to address mental health challenges faced by students in academic environments.

By integrating a robust User Dashboard and a feature-rich Admin Dashboard, the system enables personalized, real-time support through two-way communication, mood-based activity recommendations, relaxation games, and comprehensive system analytics. The enhanced Mood Tracker allows students to monitor their emotional well-being and receive tailored support, while administrators can proactively identify at-risk users and intervene with appropriate resources.

The inclusion of emergency support alerts, mood analytics, content management tools, and a game recommendation engine positions this system as a proactive mental health platform not merely a reactive one. These enhancements significantly improve user engagement, ensure personalized support, and enhance the overall effectiveness of the Mental Health Support System.

This project offers a practical and impactful real-world solution that contributes to improving student well-being and promoting mental health awareness across educational institutions in Sri Lanka and beyond.